

KEWAUNEE, WI, USA

WMO: 997741

Lat: 44.4650N

Lon: 87.4960W

Elev: 584

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -0.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 29.5	(k) 28.7	(l) 26.5	(m) 27.3	(n) 10.7	(o) 290	(p) 0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.5	(c) 82.3	(d) N/A	(e) 78.2	(f) N/A	(g) 75.0	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.5	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.0	(b) 21.2	(c) 18.9	(e) DB	(f) -10.7	(g) 88.4	(h) 6.3	(i) 3.9	(j) -15.2	(k) 91.2	(l) -18.9	(m) 93.5	(n) -22.5	(o) 95.7	(p) -27.0	(q) 98.6	(r) N/A
(4) 25.0	(5) 21.2	(6) 18.9	(7) WB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	19.0	20.3	30.5	40.9	50.5	59.2	66.2	66.2	59.1	48.9	36.7	26.3
	DBStd	18.27	11.59	10.58	9.70	6.25	6.19	6.17	5.86	4.67	6.24	6.57	8.33	9.79
	HDD50	3993	961	830	606	281	69	4	0	0	5	102	400	735
	HDD65	7949	1426	1250	1071	723	448	194	55	40	194	500	848	1200
	CDD50	1729	0	0	1	8	86	281	503	503	278	67	2	0
	CDD65	209	0	0	0	0	0	21	93	78	17	0	0	0
	CDH74	886	0	0	0	2	17	135	454	219	56	3	0	0
	CDH80	203	0	0	0	0	3	23	128	37	12	0	0	0
	WSAvg	9.0	10.3	10.2	9.9	10.5	9.4	7.3	6.8	6.9	7.7	9.1	9.8	9.9
	PrecAvg	30.7	1.5	1.1	1.8	2.8	3.4	3.9	3.6	3.1	3.0	2.6	2.1	1.7
(15) Precipitation	PrecMax	37.0	3.1	2.1	3.9	4.9	7.4	8.6	6.8	5.8	5.5	5.8	5.2	4.8
	PrecMin	25.7	0.5	0.3	0.3	0.9	1.6	1.3	1.2	0.9	1.1	0.6	0.4	0.4
	PrecStd	2.7	0.8	0.5	1.0	1.1	1.3	1.9	1.6	1.1	1.1	1.4	1.1	1.0
	PrecStd	2.7	0.8	0.5	1.0	1.1	1.3	1.9	1.6	1.1	1.1	1.4	1.1	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	44.1	56.4	64.7	77.2	84.2	89.3	84.4	80.7	70.0	56.8	47.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	40.4	38.6	49.9	56.9	68.8	78.7	83.7	79.7	74.0	63.0	51.8	43.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	37.8	36.4	46.2	51.6	63.8	73.5	79.4	76.0	70.9	60.2	49.4	40.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	35.3	34.4	42.4	48.4	59.1	68.6	75.1	73.8	68.5	58.5	47.3	38.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.2	12.8	11.0	12.5	12.0	13.4	12.5	13.2	11.4	10.2	10.8
		MCDBR	13.8	11.9	15.6	16.9	19.6	20.5	19.3	15.6	16.3	14.4	11.6	11.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.276	0.302	0.330	0.377	0.402	0.413	0.404	0.400	0.378	0.337	0.307	0.276
	taud		2.434	2.345	2.387	2.304	2.289	2.301	2.336	2.343	2.395	2.500	2.514	2.488
	Ebn at Noon		271	281	288	281	276	272	273	270	268	266	258	261
	Edh at Noon		23	30	33	39	41	41	39	37	33	26	21	20
(44) All-Sky Solar Radiation	RadAvg		466	780	1145	1417	1720	1875	1920	1640	1291	809	501	361
	RadStd		57	79	102	137	139	135	80	108	102	79	47	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page